



Artificial Intelligence Foundations (CIS4049-N)

Week 3 – Machine Learning – An Introduction – Part 2

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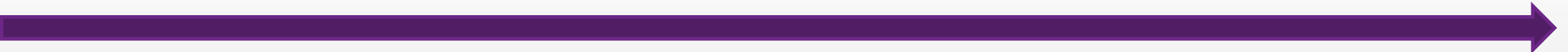
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States

Search problems
Markov decision processes
Adversarial games

Reflex

States

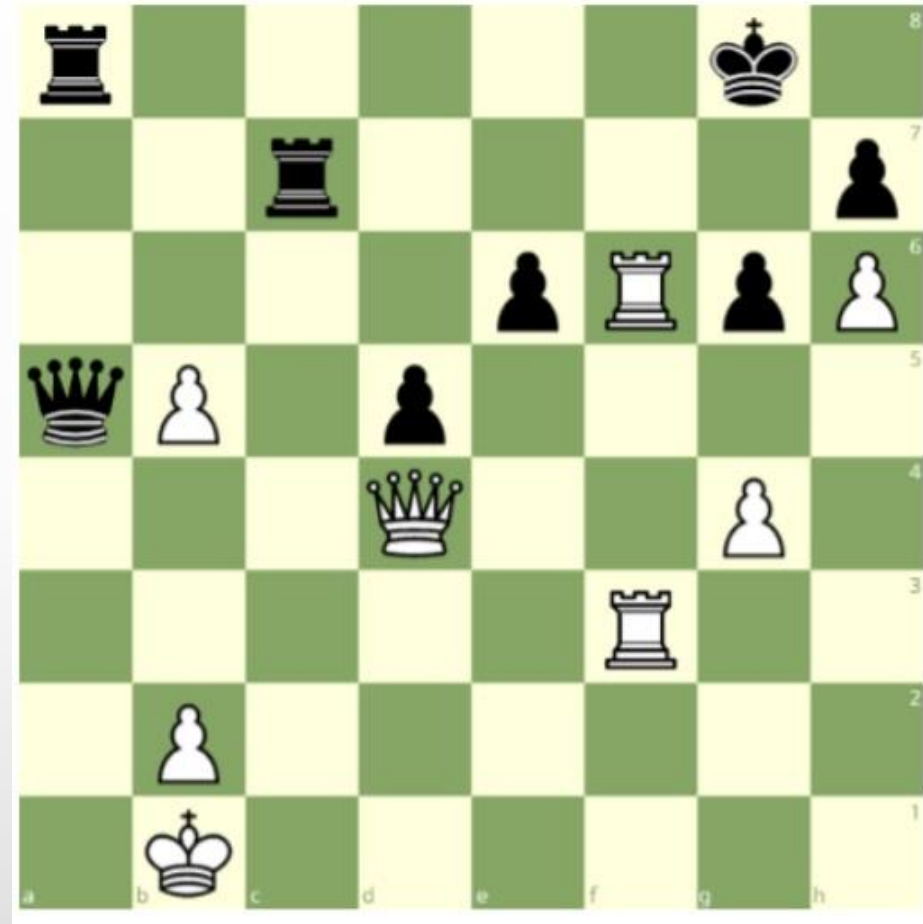


“Low-level Intelligence”

“High-level Intelligence”

Machine Learning

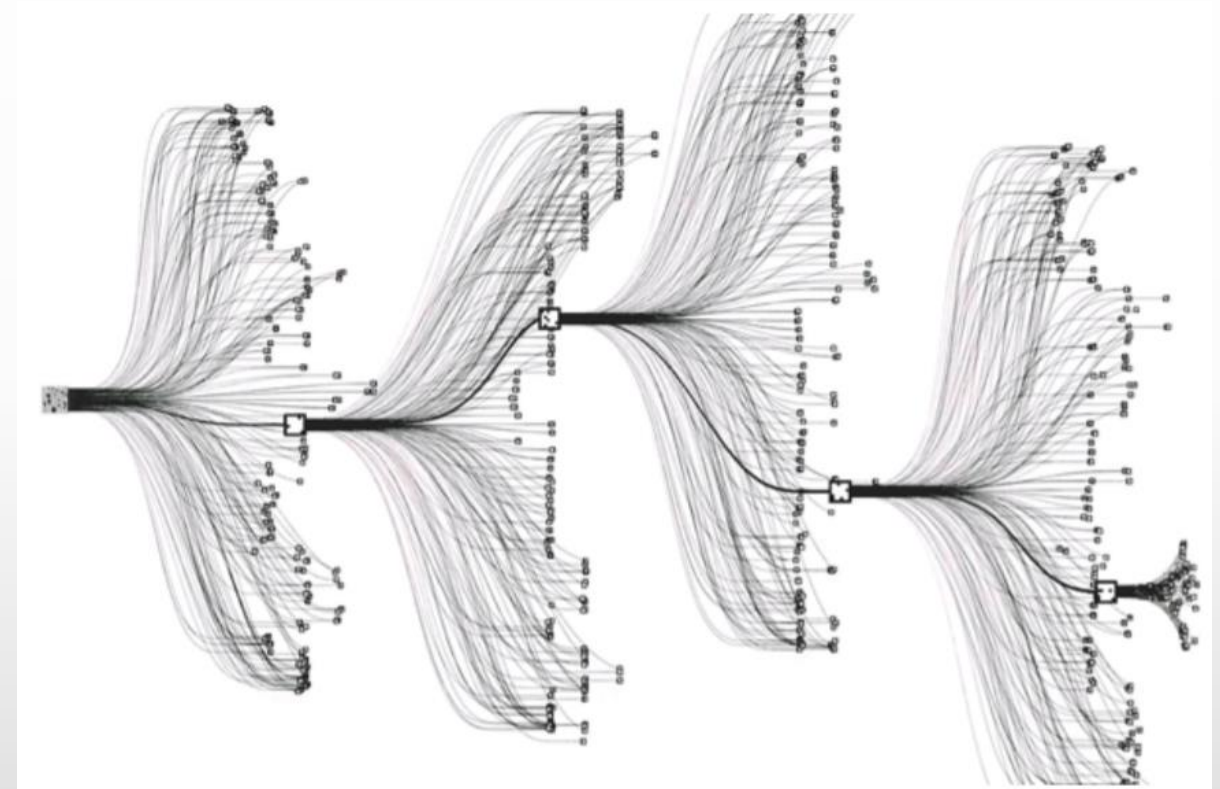
State-based Models



White to move

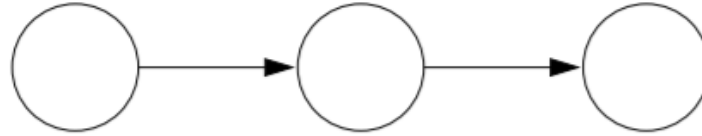
State-based Models

- Applications:
- Games: Chess, Go, Pac-Man, Starcraft, etc.
- Robotics: motion planning
- Natural language generation: machine translation, image captioning

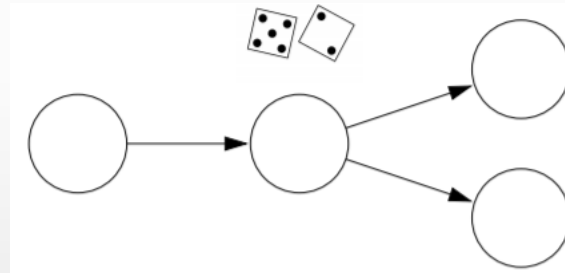


State-based Models

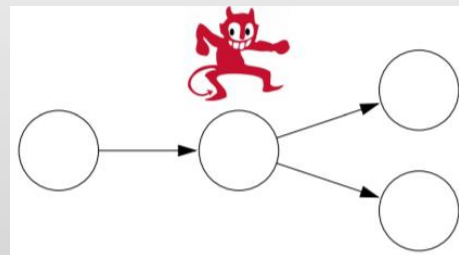
- **Search problems:** you control everything



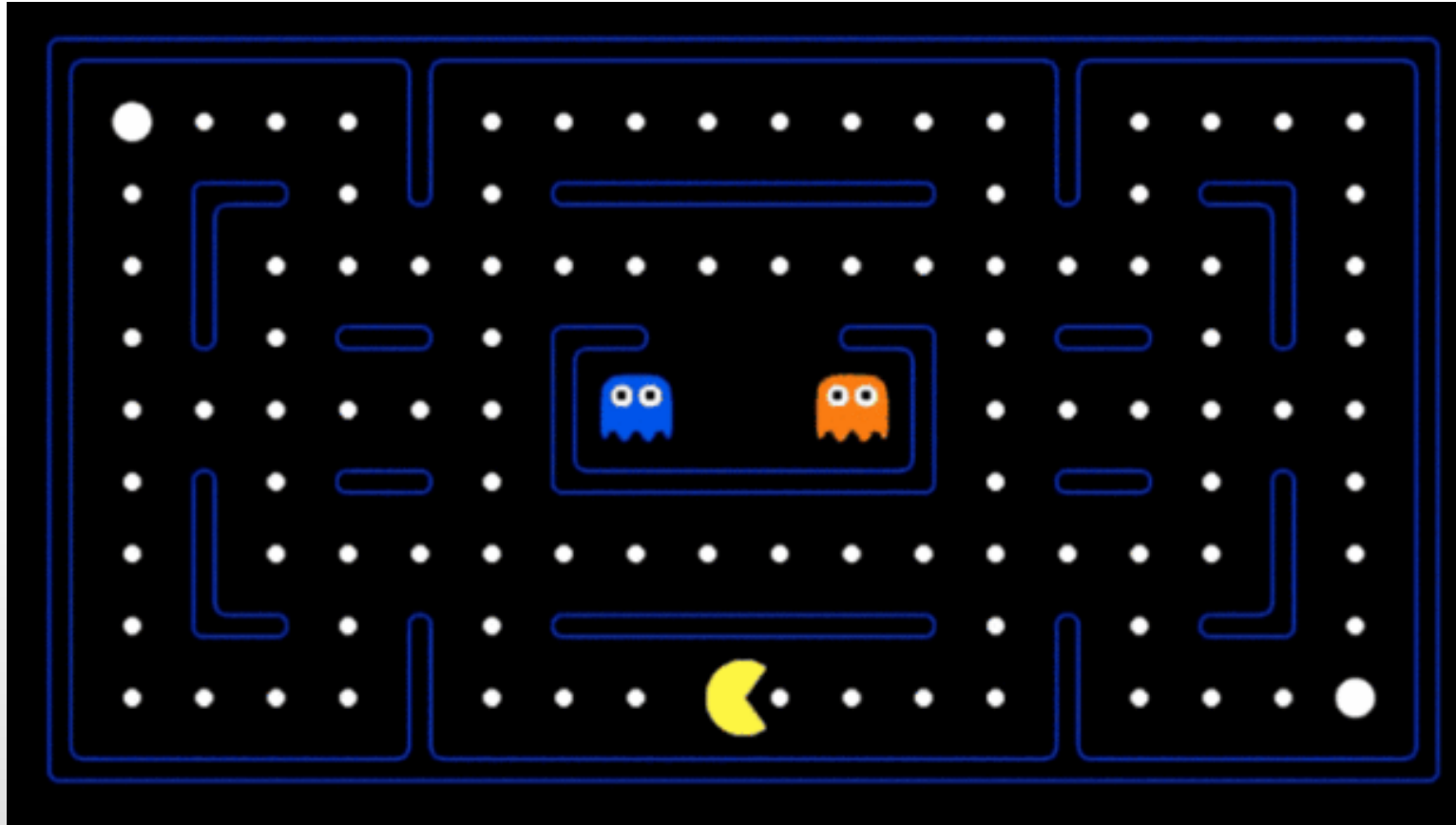
- **Markov decision processes:** against nature (e.g., Blackjack)



- **Adversarial games:** against opponent (e.g., chess)



Pac-Man



Variables

Search problems

Markov decision processes

Adversarial games

Constraint satisfaction problems

Bayesian Networks

Reflex

States

Variables

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Machine Learning

Sudoku

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9



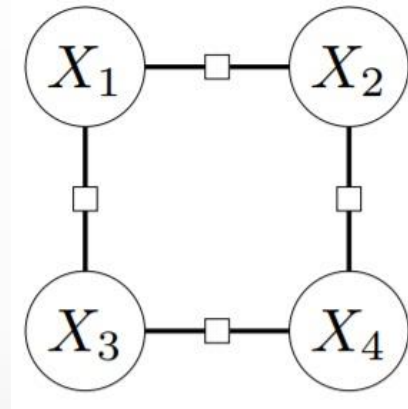
5	3	4	6	7	8	9	1	2
6	7	2	1	9	5	3	4	8
1	9	8	3	4	2	5	6	7
8	5	9	7	6	1	4	2	3
4	2	6	8	5	3	7	9	1
7	1	3	9	2	4	8	5	6
9	6	1	5	3	7	2	8	4
2	8	7	4	1	9	6	3	5
3	4	5	2	8	6	1	7	9

Goal: put digits in blank squares so each row, column, and 3x3 sub-block has digits 1–9

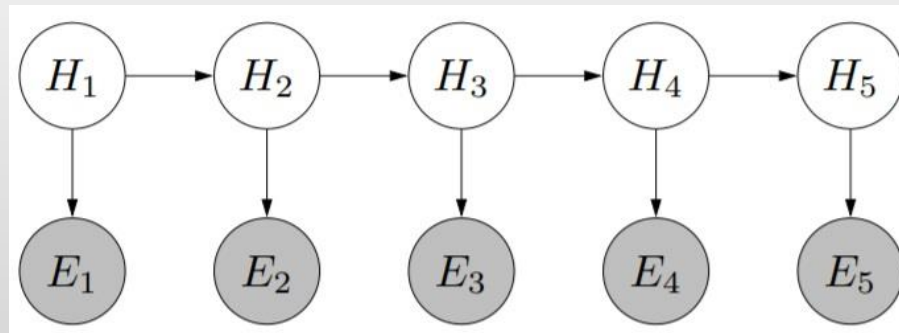
Note: order of filling squares doesn't matter in the evaluation criteria!

Variable-based Models

- **Constraint satisfaction problems:** hard constraints (e.g., Sudoku, scheduling)



- **Bayesian networks:** soft dependencies (e.g., tracking cars from sensors)



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Logic

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“High-level Intelligence”

Machine Learning

Motivation – Virtual Assistant

Tell information



Ask questions



Use natural language!

Need to:

- Digest **heterogenous** information
- Reason **deeply** with that information